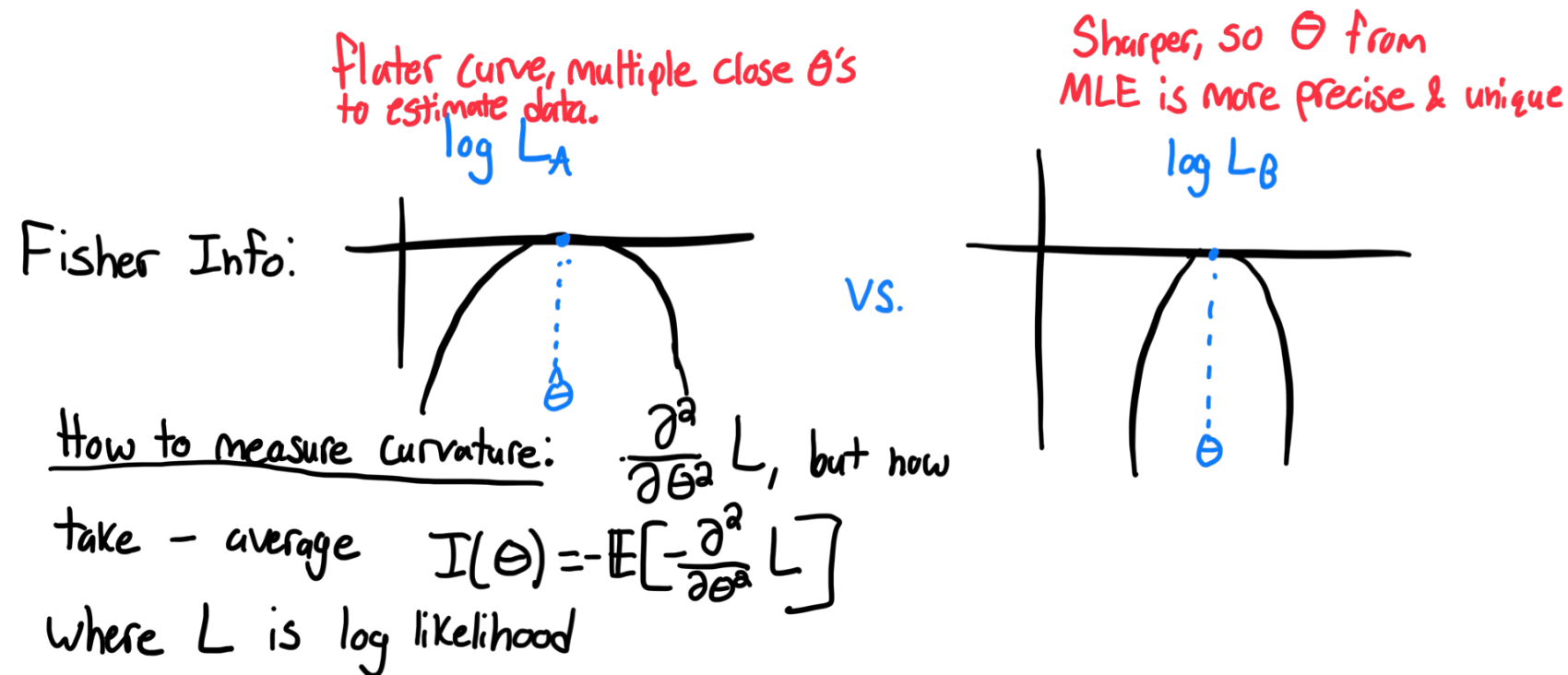




Say you have data $x_1, x_2, \dots$ and imagine it comes from a distr with parameter θ

For ex Gaussian has two param's μ and σ

Likelihood: Given data, how well does a parameter(s) explain the data.

In math terms $L(\theta) = P(\text{data} | \theta)$

Say for Bernoulli (Biased Coin). You get 7 heads in 10 flips.

Say $\theta = P(\text{heads})$. Data is i.i.d

$$L(\theta) = \prod P(x_i) = P^7 (1-P)^3$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

think of it like $P(x_i)$ happened assuming distr using θ

We do $l(\theta) = \log L(\theta)$ cuz easier math

MLE is, what is the best θ that explains the data, so $\text{MLE: } \max_{\theta} l(\theta)$

do $\frac{\partial l}{\partial \theta} = 0$ to find max

But, ... how Sharp or precise is the estimate for θ ?

Looking at graphs, we gotta measure the average curvature

Hence Fisher Info is just that $I(\theta) = -E\left[\frac{\partial^2}{\partial \theta^2} L(\theta)\right]$

Instead of assuming data samples (x_i, y_i) drawn i.i.d from $\mathcal{M}$ on $X \times Y$

Population Risk: $R(w) := E_{(x,y) \sim \mathcal{M}} [l(F(x;w), y)] := \int_{X \times Y} l(F(x;w), y) d\mu(x,y)$

$$w^* = \operatorname{argmin}_{w \in W} R(w)$$

We don't know $\mathcal{M}$ so, Empirical Risk: $\hat{R}(w) = \frac{1}{N} \sum_{i=1}^N l(F(x_i;w), y_i)$

$$\hat{w} = \operatorname{argmin}_{w \in W} \hat{R}(w)$$

Structural Risk Minimization: $\operatorname{argmin}_{w \in W} \hat{R}(w) + \lambda C(w)$

Regularization, so that model learns underlying pattern vs memorization

Ex: Tikhonov: $C(w) = \|w\|^2$ (For NN, weight decay)

Chi-Squared Distr: $Z = \mathcal{N}(0,1)$

$$Z^2 = \chi^2$$

$$V = Z^2$$

$$F(v) = P(V \leq v) = P(Z^2 \leq v) = P(-\sqrt{v} \leq Z \leq \sqrt{v}) = F_Z(\sqrt{v}) - F_Z(-\sqrt{v})$$

$$F_V(v) = F_Z(\sqrt{v}) - F_Z(-\sqrt{v})$$

$$f_V(v) = \frac{1}{2\sqrt{v}} f_Z(\sqrt{v}) + \frac{1}{2\sqrt{v}} f_Z(-\sqrt{v})$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

$$f_V(v) = \frac{1}{\sqrt{2\pi v}} e^{-\frac{v}{2}}$$

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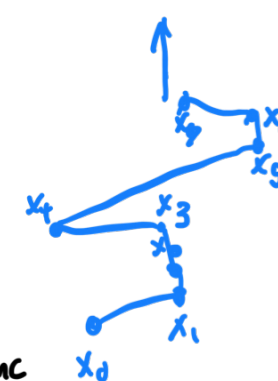
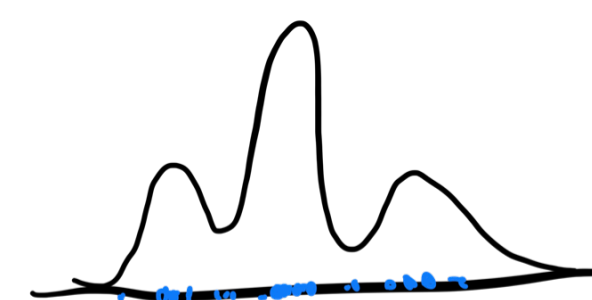
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Markov Chain Monte Carlo



Sample from dis using mcmc

Ex: Application

$$P(\theta|y) = \frac{P(y|\theta)P(\theta)}{P(y)}$$

hard to compute

$$\int P(y|\theta)P(\theta) d\theta$$

MCMC works on unnormalized distr too so doesn't matter

distr $f(x)$

take x_0 and assume its from a known distr like Gaussian then take x' from it as well

if $f(x') > f(x)$

take x' as center in "chain"

else: means center kernel around that instead

take x' with prob $\frac{f(x')}{f(x_0)}$

For infinite samples, you make distr

data = mean + residual

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

